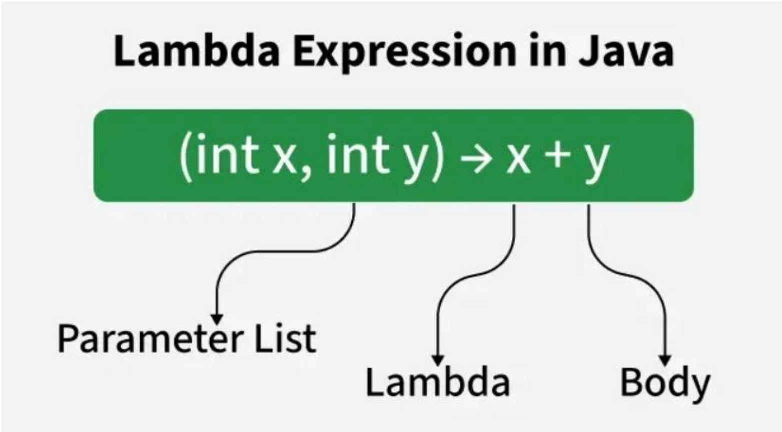
Lambda Expressions

A Lambda Expression is a short way to write anonymous functions (methods without a name). It is mainly used to implement functional interfaces.



Lambda Expression with No Parameters

```
@FunctionalInterface
interface Message {
    void display();
}

public class LambdaDemo1 {
    public static void main(String[] args) {

        Message m = () -> System.out.println("Hello Java");

        m.display();
    }
}
Hello Java
```

@FunctionalInterface is an annotation used to indicate that an interface contains exactly one abstract method.

Lambda Expression with One Parameter

```
@FunctionalInterface
interface Square {
    void findSquare(int n);
}

public class LambdaDemo2 {
    public static void main(String[] args) {
        Square s = (n) -> System.out.println("Square = " + (n * n));
        s.findSquare(5);
    }
```

```
}
Square = 25
```

Lambda Expression with Two or Multiple Parameters

```
@FunctionalInterface
interface Addition {
   void add(int a, int b);
}

public class LambdaDemo3 {
   public static void main(String[] args) {
      Addition obj = (a, b) -> System.out.println("Sum = " + (a + b));
      obj.add(10, 20);
   }
}
Sum = 30
```

```
@FunctionalInterface
interface Functional {
   int operation(int a, int b);
}

public class Test {

   public static void main(String[] args) {

      Functional add = (a, b) -> a + b;
      Functional multiply = (a, b) -> a * b;

      System.out.println(add.operation(6, 3));
      System.out.println(multiply.operation(4, 5));
   }
}
9
20
```